

Short summary/Objective: Partially validate the Multi-Gain workflow; capture ~200GB, 60k full sweeps at SC lakes; configure storage pipeline; streamline radar capture rig; test multiple HDR and decluttering methods; and prototype the proposed Masked Auto Encoder(MAE) model

Summary: Using ~200 GB of lake data, the optimal range of S1/S2 gains were narrowed to 40-90%; Furuno NXT radar features, primarily variable pulses, adjustable beam width, and dual-pulse/range modes were investigated, all while trialing prototype clutter-removal and HDR techniques on this data. Additionally, simpler hardware was implemented, reducing setup time, a parallelized CSV-to-NetCDF pipeline was developed, decreasing disk footprint, and a temporal Masked Auto Encoder prototype was produced.

Sea-ing in High-Definition: Extracting Dynamic Range in Low-Bit Sensors

Part 1: Overview of Outcomes

Table 1: Overview of Objectives & Results

Outcome	Metric/Evidence
Multi-gain workflow partially validated	Entropy analysis of 7 datasets narrowed viable S1/S2 gain values to 30-93% range.
Field data captured	≈200 GB ≈60 k full sweeps across 7 public lake sites (6 × Lake Murray + 1 × Lake Greenwood), including night-time & storm conditions.
Preliminary characterization completed	2,500 sweeps (5 ranges × 100 gains × 5 samples) logged in 40 min; results feed full-calibration design for August (20k sweeps each).
Rig & startup sequence streamlined	New rugged enclosure; Raspberry Pi boot time cut from ≥ 4 min to < 20 s after OS re-flashing and service pruning.
Storage pipeline deployed	Rust CSV-to-NetCDF converter shrank raw data footprint $\approx 25\times$ while adding richer metadata schema.
Denoising algorithms compared	Among 6 single-frame denoisers, BM3D appeared to perform best; Multi-Look median led multi-frame tests.
Early HDR fusion trialed	OpenCV's MergeMertens and NLM Algorithms improved faint-target visibility; ghosting highlighted further work.
Temporal MAE prototype trained	132 training samples; 40 epochs; masks 75 % voxels. Poor results highlight further work is needed.
Rust cleanup tool parallelized	Multi-core low-level code implementation delivers $> 4\times$ faster post-processing versus prior Python script.
Furuno NXT feature probe completed	Variable pulses, adjustable STC & dual-range modes characterized; findings guide August real-time gain scheduler.

Part 2: Gain Analysis and Collected Data Summary

Part 2a: Gain Analysis

Traditional High Dynamic Range (HDR) algorithms typically require sensor characterization to boost dynamic range. This often involves comparing a sensor's response to a "ground truth" measurement. However, when a ground truth isn't available, alternative approaches are necessary. For this system, relative gain can be calculated using various mathematical and statistical methods.

To analyze the gain, the radar was configured for minimal signal processing. Features like interpolation, RezBoost (Software-augmented Horizontal Beam Width), clutter control, and STC were all deactivated to ensure the purest possible characterization of the radar itself. Data was collected across 5 different ranges and 100 distinct gain values, with 5 samples per value. One sample set was discarded in each setting due to mid-sweep settings changes.

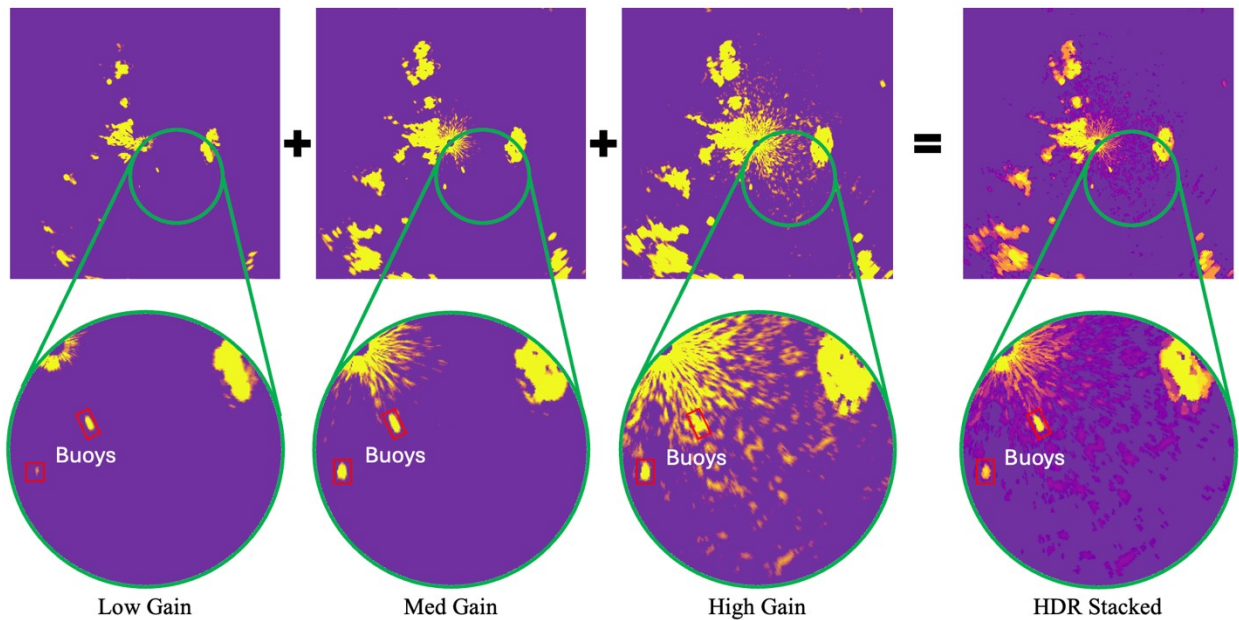


Figure 1: Example of simple HDR via Stacking Frames

Figure 1 illustrates the concept of simple HDR by stacking three 4-bit frames to effectively achieve 12 bits of dynamic range instead of 4. Notice that at high gain, distinguishing wave reflectivity from objects becomes difficult. Conversely, at low gain, some objects are barely detectable, and wave reflectivity is absent. The HDR stacked result effectively retains wave information while still enabling clear visual detection of actual objects.

One crucial method for empirically determining optimal gain values in a radar system is through the analysis of Shannon entropy. Due to the quantized nature of the radar's Analog-to-Digital Converter (ADC) and the inherent probabilistic distribution of received radar signals (comprising targets, clutter, and noise), Shannon entropy serves as a robust metric for quantifying the information content and statistical complexity of the digitized radar returns.

Shannon entropy, denoted as $H(X)$, is calculated based on the probability distribution of the discrete amplitude values (x_i) of the digitized radar signal. For a discrete random variable X with possible outcomes $\{x_1, x_2, \dots, x_n\}$ and corresponding probabilities $P(x_i)$, the entropy is given by:

$$H(X) = \sum_{i=1}^n P(x_i) \log_b P(x_i)$$

where b is the base of the logarithm (typically 2 for entropy in bits). In this context, $P(x_i)$ represents the probability of observing a specific amplitude value x_i from the digitized radar output. A higher entropy value indicates a greater degree of uncertainty, randomness, or information within the signal, whereas lower entropy suggests more predictability or less information.

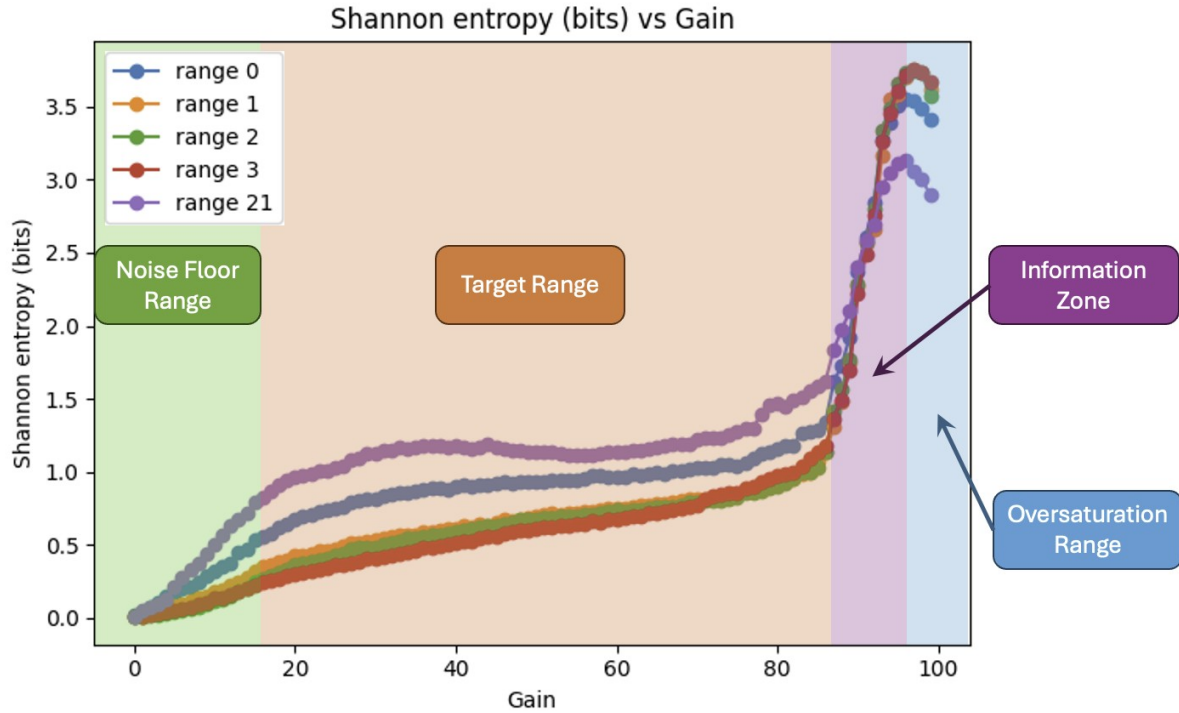


Figure 2: Entropy rates vs Gain across different range settings

The results of the Shannon entropy analysis for varying gain settings are presented in Figure 2. The observed relationship between gain and entropy can be segmented into four distinct zones, each indicating the predominant source and characteristic of the observed radar signal:

1. **Noise Floor Zone:** This initial zone is characterized by a moderate, but generally low, increase in entropy. At these minimal gain settings, the received signal is primarily dominated by the radar receiver's internal thermal noise (e.g., Johnson-Nyquist noise). While some amplification of this inherent system noise occurs, leading to a slight increase in signal variance and thus entropy, the system is largely insensitive to external targets. Any detectable objects in this zone would typically be close and possess a high Radar Cross Section (RCS). This phenomenon is evident in Figure 3.G1 and Figure 3.G15, which show that while some returns exist, it predominantly originates from areas near the radar or from targets with a high RCS.
2. **Target Acquisition Zone:** In this zone, the entropy exhibits a slow, steady increase with rising gain. This indicates the radar system is transitioning into its effective detection range. Weak target echoes, low-level ambient clutter (e.g., from ground or structures), and external environmental noise begin to be amplified above the internal noise floor. The signal's statistical complexity grows as it starts to incorporate contributions from these external sources. However, while most objects become

detectable, especially stronger ones, low-RCS targets or returns from less reflective materials like water or light rain may still be under the minimum quantized value. This can be seen in Figure 3.G20 and Figure 3.G80, where most major objects are detectable across the entire range with minimal clutter being introduced.

3. **Information Rich Zone:** This zone is marked by a dramatic increase in Shannon entropy. At these gain settings, the radar achieves 'optimal' sensitivity, ensuring that a wide range of received signals, including echoes from targets of varying RCS (even relatively low-RCS ones), significant clutter returns, and amplified noise, are well above the system's noise floor. The signal becomes highly complex and information-rich due to the superposition of these diverse and strong returns, maximizing the potential for target detection. However, it is a fine balance, as excessive gain beyond this point can lead to signal degradation. This is particularly evident in Figure 3.G85 and Figure 3.G93. Both images contain significant clutter, rain, and sea returns, indicating high information density. However, objects become more easily washed out or blend into the background at these higher gain settings compared to the previous zones.
4. **Saturation Zone:** Characterized by a sharp, often near-vertical, decrease in entropy at the highest gain settings. This drastic drop is primarily due to the saturation and clipping of the radar's Analog-to-Digital Converter (ADC) or preceding analog receiver stages. When the amplified signal exceeds the maximum voltage or digital value the hardware can process, it becomes flattened at that limit. This non-linear distortion reduces the true dynamic range of the signal, decreasing the number of unique amplitude values and thereby making the signal statistically more predictable within its clipped range, leading to a significant reduction in calculated entropy. In this zone, even moderate returns, such as those from sea clutter or heavy rain, become fully saturated, rendering the signal severely compromised and hindering effective target discrimination. As seen in Figure 3.G95 and Figure 3.G99, it's visually apparent that even sea and rain clutter are saturating the ADC, which results in reduced overall information density.

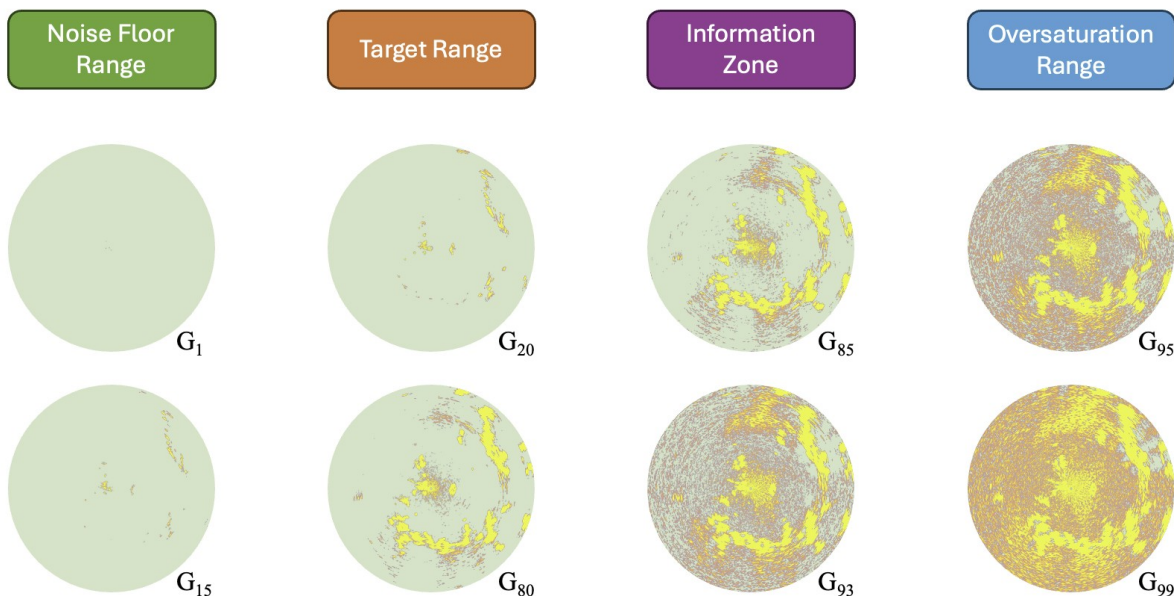


Figure 3: Different Sections of the Entropy v. Gain Chart, but shown in PPI plots

Part 2b: Field Volume & Sites

Over a focused period of 20 days, extensive radar data collection was conducted, resulting in 22 separate datasets acquired from seven strategically chosen public access points across Lake Murray and Lake Greenwood. As depicted in Figure 4, six of these collection points were located around Lake Murray, with an additional point on Lake Greenwood, each benefiting from an effective radar coverage zone extending a 1.5 nautical mile (NM) radius.

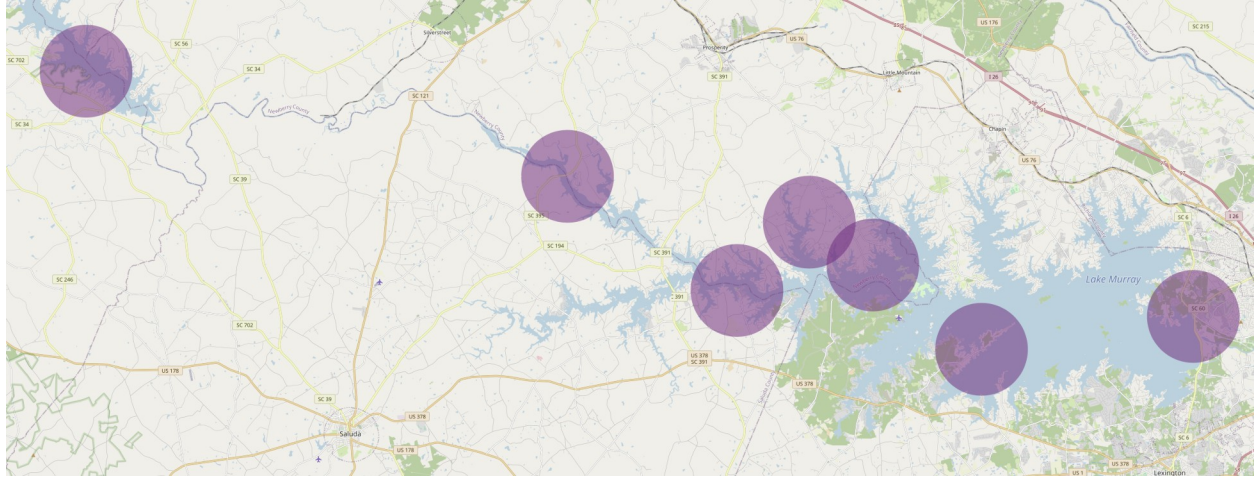


Figure 4: Radar data collection points and their effective 3 nautical mile (NM) coverage zones on Lake Murray (six points) and Lake Greenwood (one point). This extensive survey yielded approximately 60,000 radar frames, totaling over 200 GB of data.

This comprehensive survey effort culminated in the acquisition of approximately 60,000 radar frames, amounting to a substantial volume of over 200 GB of raw data. This metric directly addresses the objective of capturing a sufficient number of radar sweeps, providing a foundational dataset for subsequent analyses. The progressive accumulation of this dataset is visualized in Figure 5, illustrating the cumulative growth of radar frames from June 15th to July 4th, demonstrating consistent progress in data acquisition.

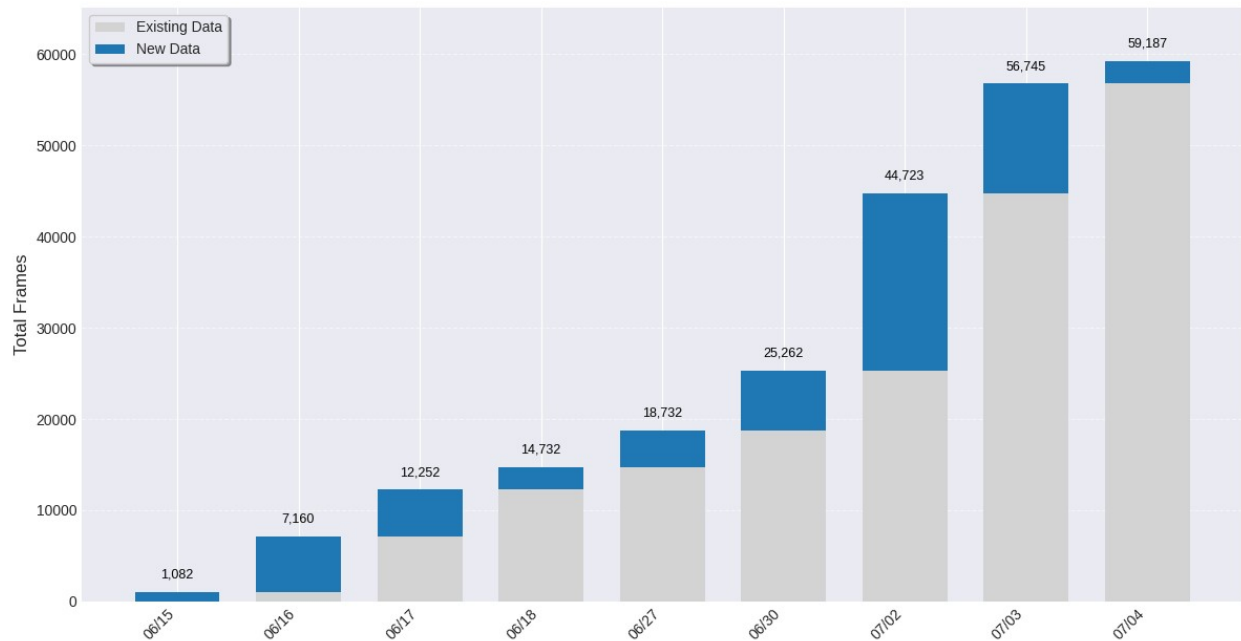


Figure 5: Cumulative growth of radar data frames over the collection period (June 15th to July 4th).

The diverse field conditions encountered during this collection period, including both nighttime and storm conditions, were crucial for robust system characterization. **Figure 6** showcases examples of nighttime data acquisition at Lake Murray, specifically highlighting setups on the north side of the dam and at the lake's upper tip where it transitions into the Saluda River.



Figure 6: Nighttime radar data collection efforts at Lake Murray. The first two images (left and center) show the field setup on the north side of the Lake Murray dam. The third image (right) captures the data collection point located at the upper tip of Lake Murray.

Complementing these nocturnal operations, **Figure 7** presents typical daytime setups across key Lake Murray locations, including Sunset Park, Shull Island Boat Ramp, and notably, the Macedonia Lutheran Church site, which proved to be the most frequently utilized collection point due to its accessibility and favorable conditions.



Figure 7: Daytime radar data collection setups at three key locations on Lake Murray. From left to right: Sunset Park, Shull Island Boat Ramp, and Macedonia Lutheran Church. The Macedonia Lutheran Church site (right) was the most frequently utilized data collection point throughout the survey.

The variety of these environmental settings and geographical locations ensures that the collected data encompasses a broad spectrum of real-world scenarios, which is vital for validating the Multi-Gain workflow. Specifically, the field data allowed for the entropy analysis (Part 2a), which subsequently narrowed the viable S1/S2 gain values to a 30-93% range based on the information-rich zone findings (Figure 2 and Figure 3). This extensive field data collection and site diversity were paramount in laying the groundwork for optimizing the radar's performance characteristics for High Dynamic Range applications.

Part 3: Furuno Radar Details & ARPA

Radar systems, like the Furuno unit(s) used in this study, operate by emitting electromagnetic pulses and interpreting the reflected signals. The precision and utility of the data derived from these systems are heavily influenced by the fundamental parameters of these pulses and the subsequent signal processing. The ability to manipulate and understand these parameters is crucial for achieving high dynamic range (HDR) and effective clutter removal, which are some of the goals of this research.

Part 3a: FURUNO specs

The core of radar operation involves the transmission of energy in discrete bursts and the measurement of returns. For the Furuno system in use, a single radial (a line extending outwards from the radar's center) is composed of 868 'dots' or range cells, representing individual range measurements. A full 360-degree sweep is captured by approximately 600 radials in this system with 'high RPM'. However, with advanced functionalities such as Dual Pulse, this data density can effectively be doubled to ~1200 radials per sweep, providing a significantly finer resolution

of the surrounding area. This dense data capture is critical for later computational analysis and the application of machine learning models like the proposed MAE.

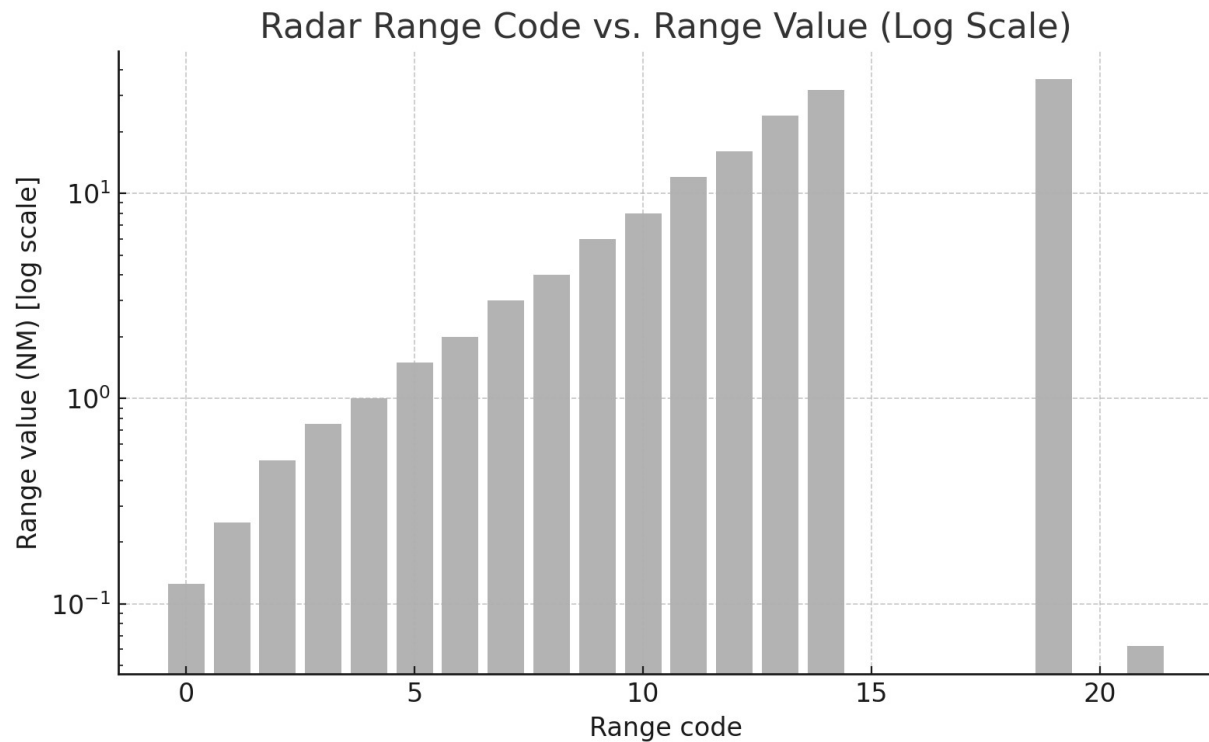


Figure 8: Log-scaled representation of radar range value (NM) versus range code, illustrating possible range settings from 0.0625 NM to 36 NM.

Figure 8 demonstrates the vast operational range of the radar, from very close-quarters (0.0625 NM) to extended offshore distances (36 NM). The log-scaled representation highlights the non-linear relationship between the internal range code and the actual nautical mile value. This non-linearity necessitates a robust understanding of the radar's response across its entire operational envelope, particularly when applying algorithms designed to extract higher dynamic range. The ability to select appropriate ranges for data collection was fundamental to the field data capture summarized in Part 2b.

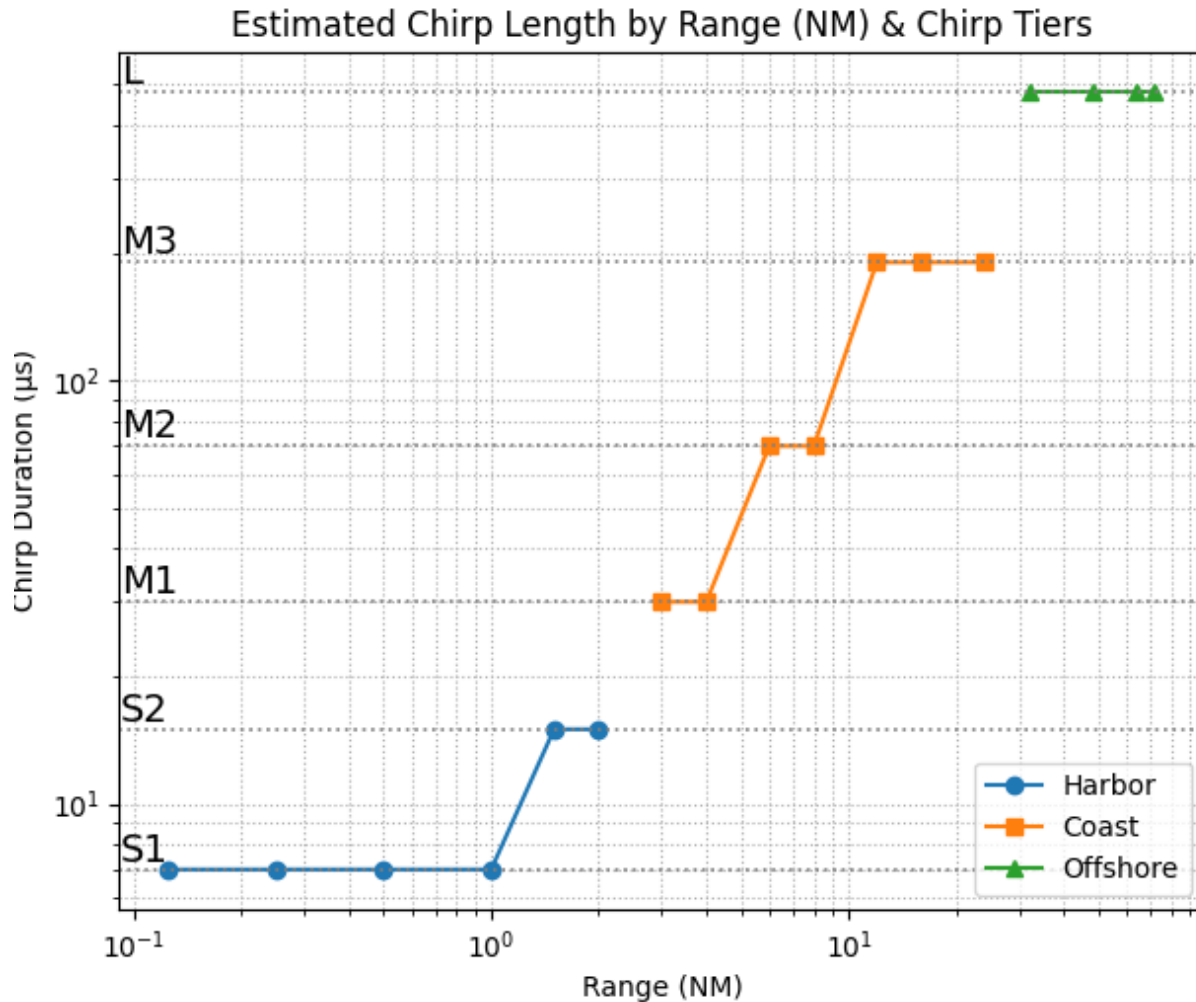


Figure 9: log-log plot illustrates the estimated chirp duration (pulse length in microseconds) as a function of radar range (nautical miles), specifically highlighting different operational tiers. The S1 and S2 designations represent shorter pulse lengths primarily suited for close-range (Harbor) detection, M1, M2, and M3 denote medium pulse lengths for coastal operations, while the 'L' tier signifies long pulse lengths optimized for extended offshore ranges.

The choice of pulse length is critical for radar performance. As illustrated in Figure 9, Furuno radars utilize various "chirp durations" or pulse lengths, each optimized for specific operational tiers.

Shorter pulse lengths (S1, S2 - Harbor tiers) are ideal for close-range detection, offering high range resolution to distinguish between closely spaced targets like buoys or other vessels in congested waterways. The trade-off is typically lower overall power and thus shorter detection range.

Medium pulse lengths (M1, M2, M3 - Coastal tiers) provide a balance between range resolution and detection capability, suitable for navigating coastal areas where both detail and moderate range are required.

Long pulse lengths (L - Offshore tiers) carry more energy, allowing for greater detection range, crucial for identifying distant targets in open waters. However, longer pulses inherently reduce range resolution, meaning closely spaced targets at long range might appear as a single, elongated echo.

The ability to vary pulses by adjusting their length is a fundamental aspect of optimizing radar performance for different scenarios. A shorter pulse provides better discrimination between targets at close range, while a longer pulse increases the total energy transmitted, enhancing detection of weaker targets at greater distances. For this research, understanding these pulse characteristics is vital for developing effective HDR and decluttering techniques, as the optimal pulse for capturing information in one zone (e.g., harbor) may be entirely different from another (e.g., offshore).

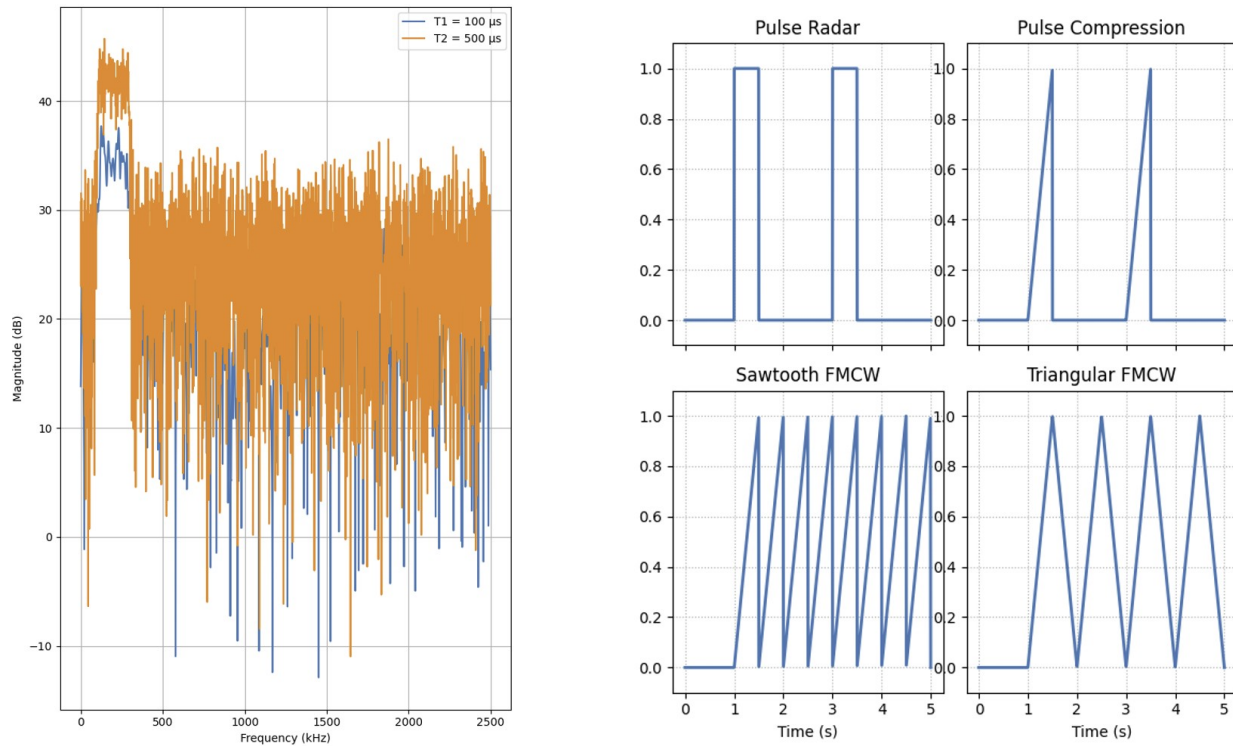


Figure 10: (Left) Comparison of signal magnitude versus frequency for two pulse durations ($100 \mu s$ and $500 \mu s$), demonstrating the improved signal parsing with a longer pulse. (Right) Visual representation of different radar waveforms: Pulse Radar, Pulse Compression, Sawtooth FMCW, and Triangular FMCW.

Figure 10(Left) underscores the impact of pulse duration on signal quality. A longer pulse, such as $500 \mu s$, can allow for "improved signal parsing" or more detailed analysis of the return signal, potentially leading to better target identification and tracking, especially when coupled with advanced processing techniques like pulse compression. Figure 10(Right) provides insight into various radar waveforms, beyond simple pulse radar. A comparison of waveforms is provided in Table 2.

Table 2: Comparison between Different Radar Waveforms

	Description	Major Benefits	Key Limitations
Conventional Pulse	Transmits short, high-power electromagnetic pulses and listens for echoes. Range is determined by measuring the time delay between transmission and reception.	Simplicity of design and operation. Good range resolution with very short pulses.	Direct trade-off between range resolution (short pulse) and detection range (long pulse). Less detection energy with short pulses.
Pulse Compression	Transmits a relatively long, internally modulated pulse (e.g., chirped frequency), which is then digitally "compressed" upon reception into a much narrower, high-amplitude pulse through signal processing (e.g., matched filtering).	Achieves high energy (long range) and high resolution (short effective pulse). Improved Signal-to-Noise Ratio (SNR). Enhanced clutter rejection due to unique spectral properties.	Requires more complex signal processing on reception.
Sawtooth FMCW	Transmits a continuous wave whose frequency changes linearly in a "sawtooth" pattern. Range is calculated by measuring the frequency difference between the simultaneously transmitted and received signals.	Very high range resolution (centimeter-level) at short ranges. Low transmitted peak power (inherently safer, energy efficient). Can provide Doppler (velocity) information.	Primarily effective at shorter ranges. Susceptible to interference from other FMCW radars.
Triangular FMCW	Similar to Sawtooth FMCW, but the transmitted wave's frequency sweeps both up and down in a triangular pattern. Range and velocity are determined from frequency differences observed during both the up-sweep and down-sweep phases.	All benefits of Sawtooth FMCW. More robust and unambiguous measurement of both range and velocity due to bidirectional sweep, helping to resolve potential ambiguities.	Similar range limitations to Sawtooth FMCW. Still primarily for shorter-range applications.

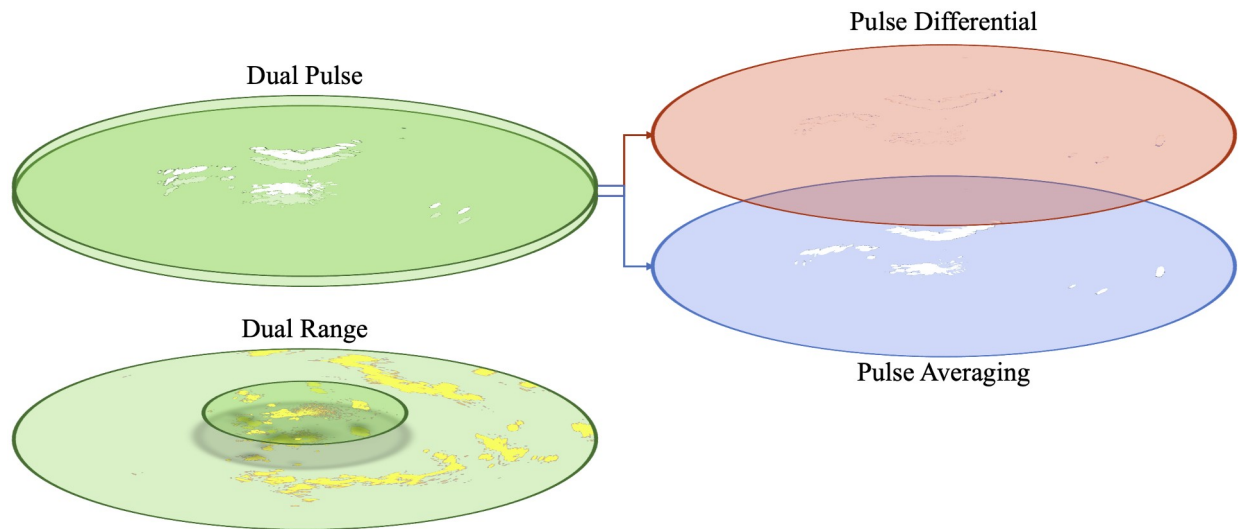


Figure 11: Comparison of Dual Pulse and Dual Range radar functionalities. Dual Pulse (*top*) offers improved interpolation and signal clarity by utilizing Pulse Differential (*red*), which detects movement and effectively narrows the horizontal beam, and Pulse Averaging (*blue*), which filters out fast-moving noise. Dual Range (*bottom*) enhances spatial resolution, providing a clearer view for targets at both close and far distances simultaneously.

Figure 11 highlights two significant enhancements to radar performance directly relevant to this report: Dual Pulse and Dual Range. The advanced Dual Pulse feature refines radar return quality for improved signal clarity, crucial for HDR applications. This is achieved via Pulse Differential, which compares successive returns to detect target movement, sharpening the horizontal beam by distinguishing moving targets from stationary clutter and clarifying vessel traffic. Concurrently, Pulse Averaging reduces random noise by averaging multiple returns, enhancing the signal-to-noise ratio to discern weaker targets amidst interference like rough seas or rain. This dual functionality effectively doubles the number of radials per sweep (from ~600 to ~1200), enabling more detailed spatial data capture. Complementing this, Dual Range simultaneously displays two different range scales in addition to gain, offering both broad situational awareness and fine-detail navigation without constant switching. For HDR, this provides critical multi-exposure information from a single scan, supporting later HDR work.

Part 3b: Furuno Data Augmentation

The Furuno SDK and platform offer a comprehensive suite of software features designed to enhance radar performance and operator interpretation. Among these, RezBoost (Effective Azimuth Beam Width) is prominently touted for its ability to provide 'clearer' data, ostensibly making it easier for an operator to discern real objects from clutter on the display. This function claims to significantly reduce the effective horizontal beam width from a physical 3.8 degrees down to approximately 1.9 degrees. The visual impact of RezBoost is compellingly demonstrated in **Figure 12**, where the radar image processed with RezBoost (left panel) exhibits sharper, more defined returns, appearing less spread out than the 'Standard' setting (right panel). This sharpening is achieved by sophisticated digital signal processing that simulates the effect of a narrower physical antenna beam. While this undoubtedly benefits human operators by reducing angular spreading and improving the visual separation of closely spaced targets, its utility for the computational research undertaken here is likely limited. There is a concern that RezBoost may operate by applying internal filters or transformations that could potentially discard or smooth out valuable raw data, particularly the subtle variations critical for high dynamic range (HDR) extraction or the nuanced features required by machine learning models like the Masked Auto Encoder (MAE). For this research, maintaining access to the purest, least-processed data is ideal.

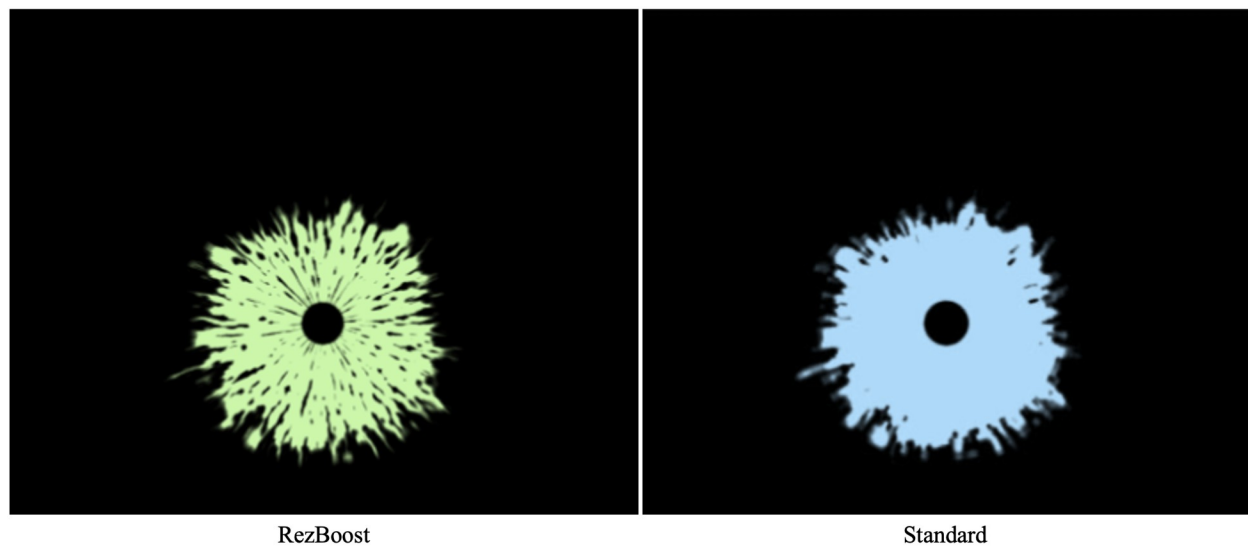


Figure 12: RezBoost Max(3) vs Standard Settings

Beyond visual enhancements, the Furuno system also employs intelligent data handling, particularly concerning interpolation and its interaction with the Dual Pulse functionality. Despite claims of being an 8-bit system, raw data collection consistently reveals only 17 distinct amplitude values, effectively rendering it a 4-bit system with one additional level. This limited quantization presents a notable challenge for extracting high dynamic range. However, when interpolation is activated, the system provides a full 8-bit output while concurrently removing the Dual Pulse functionality and providing additional radials. This suggests the presence of an internal algorithm generating synthetic or interpolated data points to create a smoother transition in results and to fill in the perceived missing dynamic range.

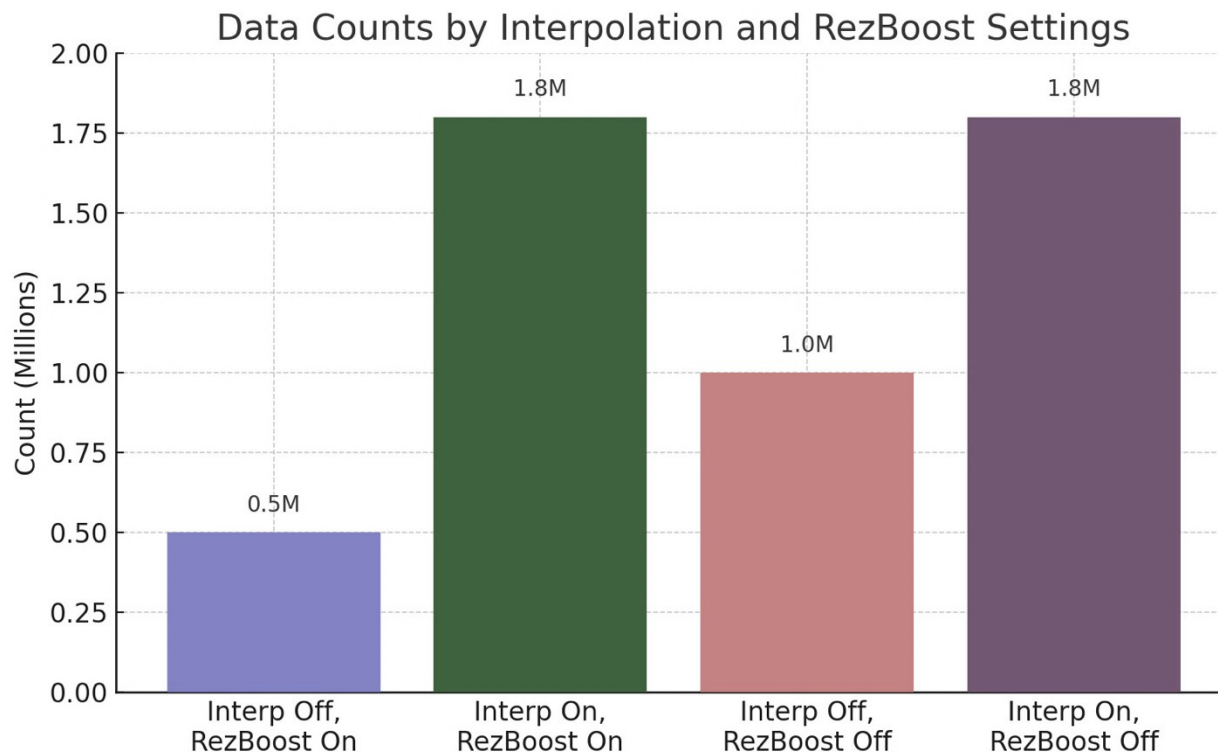


Figure 13: Interpolation and RezBoost effect on data usage

The impact of these settings on data acquisition volume is quantitatively presented in **Figure 13**. This bar chart illustrates how enabling or disabling Interpolation and RezBoost affects the number of data points (in millions) captured per sweep. For instance, with "Interp Off, RezBoost On," data counts are approximately 0.5 million, suggesting that while RezBoost provides visual enhancement, it does not increase raw data density without interpolation. Conversely, the "Interp On, RezBoost On" configuration yields the highest data count at approximately 1.8 million, indicating that interpolation, even when coupled with RezBoost, significantly increases the volume of reported data, likely incorporating the additional radials and interpolated values. The "Interp Off, RezBoost Off" setting, representing a more raw data capture with limited bit depth, still provides a substantial 1.0 million data points, aligning with the expected density when Dual Pulse is active. Finally, the "Interp On, RezBoost Off" scenario also registers approximately 1.8 million data points, identical to the "Interp On, RezBoost On" case, which further supports the hypothesis that the act of interpolation itself is the primary driver for the increased data points, irrespective of RezBoost's state.

Part 3c: ARPA vs mARPA & Furuno Implementation

Radar systems for collision avoidance are broadly categorized into ARPA (Automatic Radar Plotting Aid) and mARPA (mini-ARPA), distinguished by their automation and target tracking capacity. Full ARPA systems, typically found on larger commercial vessels, automatically detect, acquire, and track targets, calculating critical collision parameters (CPA, TCPA) and providing comprehensive target data (ID, bearing, range, COG, SOG). These systems require precise heading and speed inputs, powerful processing (e.g., Kalman filtering), and a high-resolution radar front-end. In contrast, mARPA, common

on smaller recreational craft, requires manual target selection, tracks fewer targets (10-30), has slower updates, lacks automatic hazard detection, and may occasionally drop tracks. While still valuable, mARPA's limitations make it a less sophisticated collision avoidance tool. A more complete summary is seen in Table 3.

The Furuno DRS4D-NXT radar, used in this project, possesses full ARPA capabilities, a significant feature for a leisure-class solid-state Doppler radar. It automatically tracks up to 100 targets, providing rapid speed and course vectors and detecting low-RCS objects. The ability of the Furuno NXT to generate comprehensive tracking data, even when internal processing is bypassed for raw data capture, provides a rich dataset for supervised training. The presence of full ARPA may also serve as a benchmark for evaluating the performance of the algorithms developed within this study. Currently, there has been no success in implementing ARPA/mARPA across any of the Furuno & Simrad radars in this project.

Table 3: Summary of ARPA and mARPA capabilities

Feature	ARPA (Full)	mARPA (Mini)
Target Acquisition	Automatic (self-plots based on Doppler/guard-zone rules) and manual	Manual only (operator must select each echo)
Track Updates	Every antenna rotation (≤ 2.5 s @ 48 RPM)	Slower cadence (often tied to slower processor/sweep rates)
Motion Vectors	True- and relative-motion vectors drawn for every track	Often only basic relative vectors; may lack true-motion computation
Collision Alarms	Continuous CPA/TCPA calculation; audible/visual alarms on threshold breach	No automatic hazard detection or alarm generation
Maximum Tracks	Up to 100 targets (modern sets)	Typically 10–30 targets
Track Robustness	Maintains tracks through course/speed changes; minimal drop-out	May drop targets when echoes weaken or vessel maneuvers
Platform Size / Cost	Larger, commercial vessels; requires high-performance processor and sensor inputs	Smaller recreational craft; lower-cost, less-powerful hardware
Supplemental Needs	Gyro-compass ($\leq 0.5^\circ$ error), speed input (log/GPS), high-res pulse-compression radar, Kalman filter	No gyro/speed inputs required; simpler radar front-end and correlation logic
Typical Use Case	Primary collision-avoidance on commercial ships, offshore support, naval vessels	Supplementary collision-awareness on yachts, fishing boats, small workboats

Part 4: Decluttering and Dynamic Range Algorithm Analysis

In maritime radar imaging, two of the most formidable challenges are (1) clutter and speckle noise, which obscure real targets against a highly variable sea return, and (2) the wide dynamic range of reflectivities, from faint buoys and small craft to large steel hulls, that must be rendered simultaneously. Effective decluttering enhances detection and tracking performance, while extended dynamic-range processing ensures that both low- and high-intensity echoes are visible in a single composite.

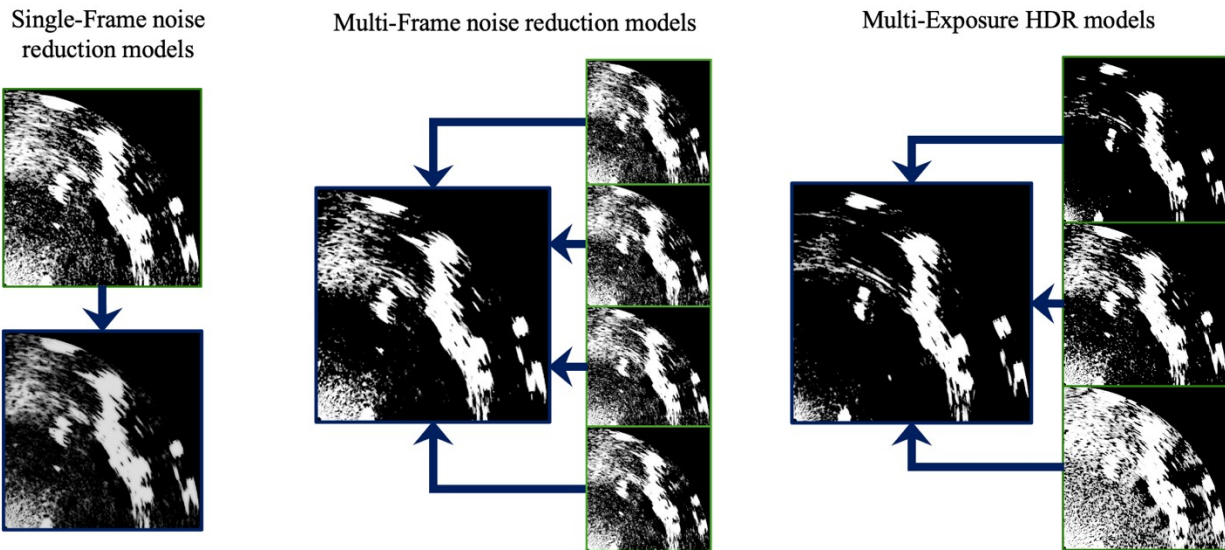


Figure 14: Overview of the three primary algorithms tested

To address these challenges, three broad classes of algorithms were evaluated, as shown in Figure 14.

Single-Frame Noise-Reduction Models operate on each radar sweep independently, estimating local statistics such as mean, variance and patch similarity to distinguish noise from signal. Methods like Lee filtering, wavelet denoising and BM3D adaptively smooth homogeneous regions while preserving edges. Because they require no temporal alignment, these approaches offer the lowest latency and simplest integration, but can only exploit spatial structure and typically blur small or low-contrast targets.

Multi-Frame Noise-Reduction Models leverage a short sequence of consecutive sweeps, exploiting the fact that true clutter (sea surface, stationary objects) remains correlated over time while speckle and random noise do not. Techniques ranging from pixel-wise median averaging and robust principal component analysis (RPCA) to non-local means across frames achieve significantly stronger noise suppression, but at the expense of sensitivity to moving targets and the need for robust registration.

Multi-Exposure (HDR) Fusion Models combine radar returns captured at different gain or exposure settings, which encode complementary information: high-gain sweeps reveal faint echoes but saturate strong targets, while low-gain sweeps preserve bright returns but lose weak signals. HDR fusion algorithms (for example, exposure-fusion or the MergeMertens method) automatically select and blend the best-exposed regions from each image, producing a single radar map that simultaneously displays both ends of the intensity spectrum. This approach excels at extending dynamic range without explicit sensor calibration, though it requires careful alignment to avoid ghosting and can blend in noise if the exposures are poorly matched.

Part 4a: Single-frame denoising

Single-frame denoising techniques operate on each radar sweep individually, without leveraging temporal correlations. Figure 15 showcases visual results from four representative algorithms tested under consistent conditions. These techniques vary in their mathematical assumptions, filtering domains, and trade-offs between noise suppression and target preservation.

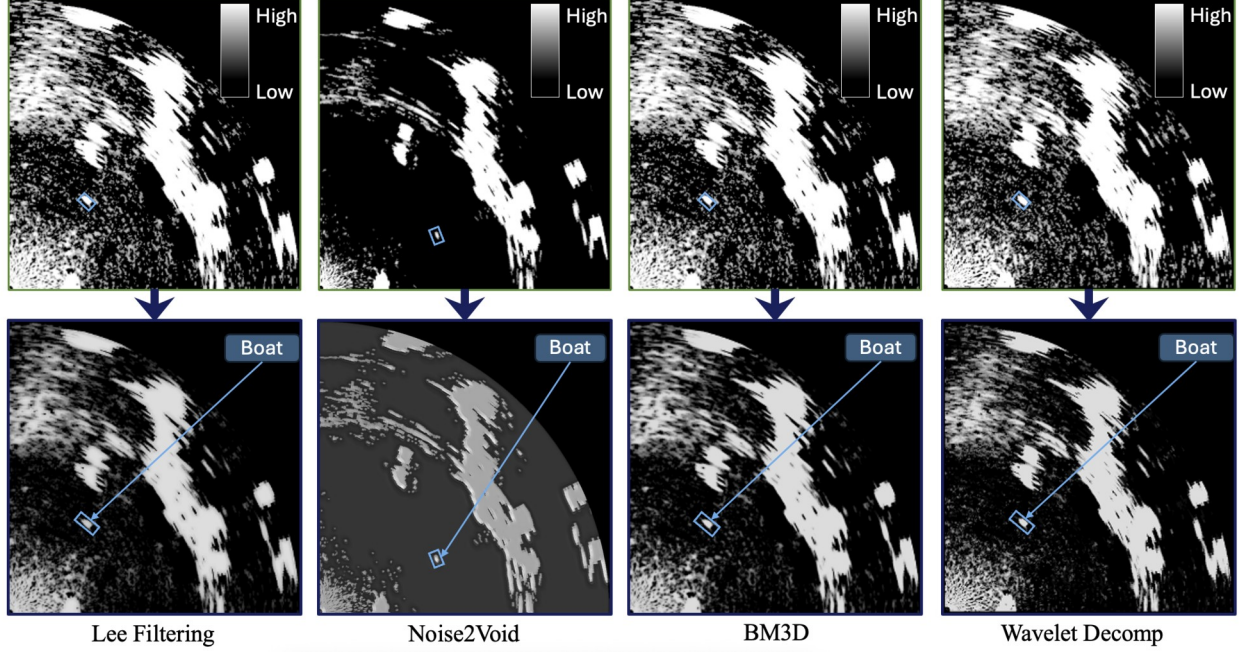


Figure 15: A Visual comparison of Lee Filtering, Noise2Void, Bm3D, and Wavelet Decomposition

Lee filtering assumes a multiplicative speckle noise model commonly observed in coherent radar systems. The observed image intensity I is modeled as $I = S \cdot N$, where S is the true signal and N is the multiplicative noise. Within a local window Ω , the filtered output is computed as a linear blend of the local mean μ_Ω and the observed value:

$$I_{\text{filtered}} = \mu_\Omega + W \cdot (I - \mu_\Omega),$$

with weight

$$W = \frac{\sigma_\Omega^2 - \sigma_n^2}{\sigma_\Omega^2},$$

where σ_Ω^2 is the local variance and σ_n^2 is the noise variance. This filter is edge-aware: where local variance is high (i.e., near object boundaries), smoothing is reduced. However, in practice, Lee filtering tended to over-smooth returns near point targets and failed to suppress higher-order speckle noise effectively, leading to residual clutter seen in Figure 15.

Noise2Void uses a self-supervised convolutional neural network trained to predict the intensity of masked pixels from their surroundings. It leverages a U-Net architecture with blind-spot convolutions to avoid identity learning. The objective function minimized during training is

$$L = \sum_{i \in M} \|f_\theta(I_{\setminus i}) - I_i\|_2^2,$$

where M is the set of masked pixels, and f_θ is the CNN output. Since training does not require clean targets, it is adaptable to arbitrary radar speckle profiles. Despite its adaptability, in our radar scenes it

smoothed out low-RCS (Radar Cross Section) targets and failed to sufficiently suppress coherent noise near edges, as seen in the comparative patches of Figure 15.

BM3D(Block-Matching and 3D Filtering) is a two-stage, non-local transform-domain algorithm. It begins by grouping similar patches P_i from the image into 3D arrays via block matching using Euclidean norms. The first stage applies a 3D linear transform (typically DCT or Haar), followed by hard thresholding in the coefficient domain:

$$\hat{C} = T^{-1} \left(H \left(T(P) \right) \right)$$

where T is the 3D transform and H is a hard-threshold operator. In the second stage, patches are aggregated around the pre-filtered estimate and processed via collaborative Wiener filtering:

$$\hat{P}_{wiener} = T^{-1} \left(W \cdot T(P) \right)$$

with W computed from estimated signal variance. BM3D outperformed other single-frame methods by balancing noise suppression and retention of target edges and structures, making it a very good single-frame method in Figure 15.

Wavelet Decomposition (Multiresolution Denoising) approach performs a discrete wavelet transform (DWT) to decompose the image into sub-bands across multiple spatial resolutions. Let I be the radar intensity image; then

$$I = A_j + \sum_{k=1}^j D_k$$

where A_j is the approximation at scale j , and D_k are detail coefficients. Denoising is applied via shrinkage:

- a) **Hard thresholding:** $D'_k = D_k \cdot 1_{|D_k| > T}$
- b) **Soft thresholding:** $D'_k = \text{sign}(D_k) \cdot \max(|D_k| - T, 0)$

The choice of threshold T can follow universal rules (VisuShrink) or Bayesian models (BayesShrink). In the results, wavelet methods preserved large target structures but often obliterated weaker returns and introduced minor block artifacts due to hard coefficient pruning, as seen in Figure 15.

Part 4b: Multi-frame denoising

Multi-frame denoising algorithms exploit temporal correlations across successive radar sweeps. These methods assume that noise is random and uncorrelated between frames, while persistent clutter or real targets exhibit temporal coherence.

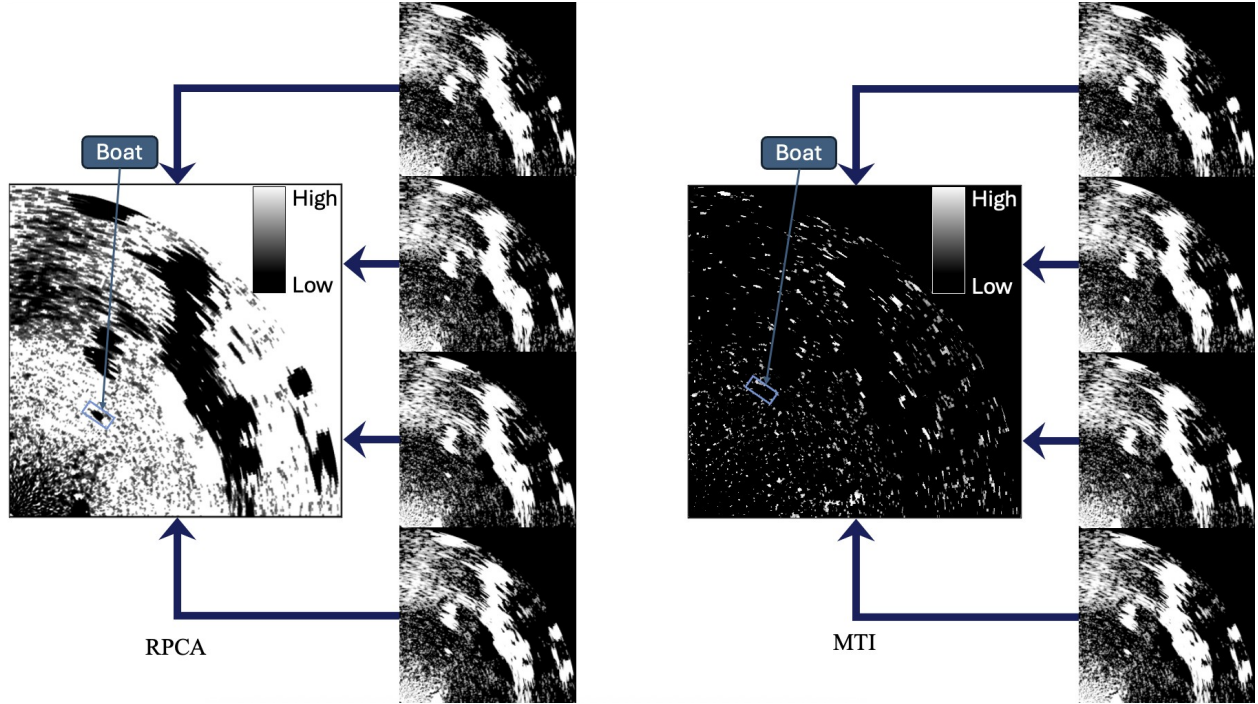


Figure 16: A visual Comparison of RPCA and MTI. These algorithms performed worst among all algorithms tested across the 3 classes

RPCA(Robust Principal Component Analysis) seeks to decompose a temporal stack $D \in R^{m \times n}$ (where each column is a vectorized radar frame) into a low-rank matrix L (background) and a sparse matrix (foreground and noise) via:

$$\min_{L, S} ||L||_* + \lambda ||S||_1 \text{ s.t. } D = L + S$$

Here, $||\cdot||_*$ denotes the nuclear norm (sum of singular values), promoting low-rank structure and $||\cdot||_1$ promotes sparsity. Optimization is typically performed using the inexact augmented Lagrange multiplier method. RPCA strongly suppressed random clutter but erroneously pushed stationary objects (e.g., anchored buoys) into the background, making them nearly invisible (Figure 16). Computational cost also scales poorly with frame size and sequence length.

MTI (Moving Target Indication) exploits Doppler processing principles by subtracting frames or applying filter banks tuned to velocity. A simple two-frame canceller is:

$$y[n] = x[n] - x[n-1]$$

which cancels static or slow clutter. However, MTI performance depends critically on the radar PRF, sweep rate, and accurate alignment. In our case, azimuth jitter and platform motion caused strong “false movement” artifacts. MTI failed to isolate real movers (targets) and instead highlighted edge noise and frame misalignments, as shown in Figure 16.

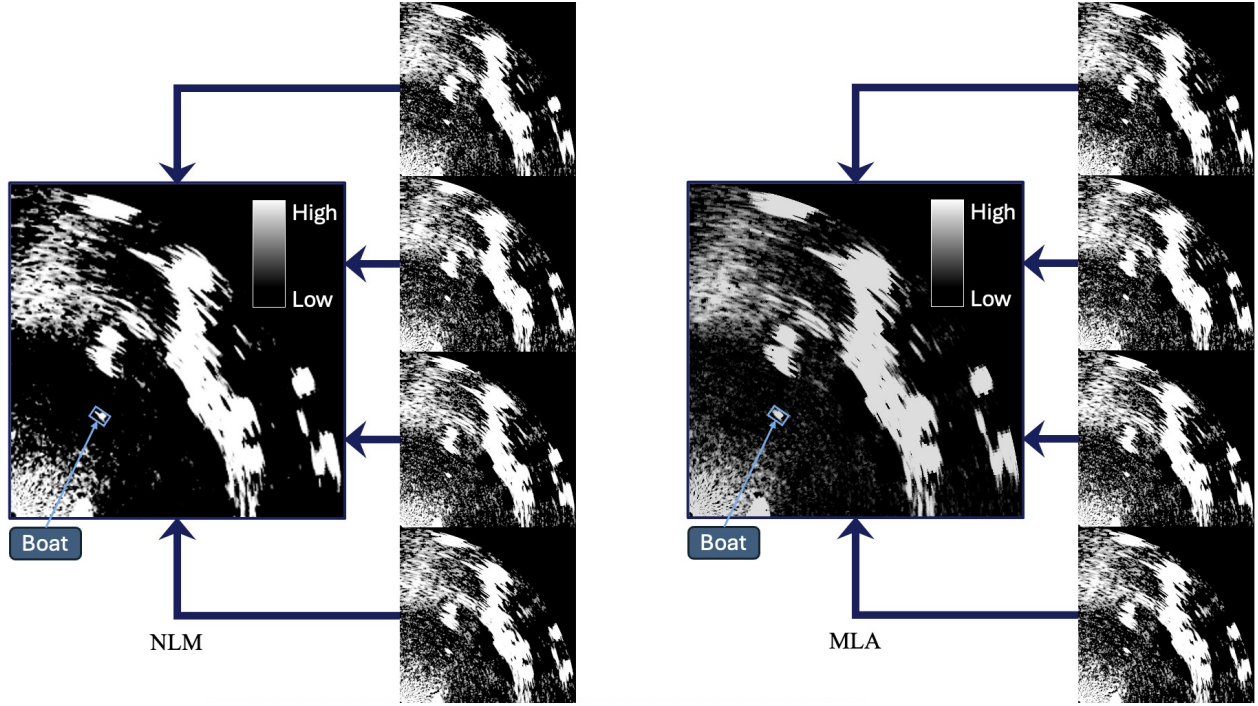


Figure 17: A visual comparison of NLM and MLA. NLM appeared to perform best.

MLA (Multi-Look Averaging) reduces speckle variance by averaging across N co-registered frames:

a) Mean estimator: $I_{\text{MLA}} = \frac{1}{N} \sum_{i=1}^N I_i$

b) Median estimator (more robust to outliers): $I_{\text{MLA}} = \text{median}(I_1, I_2, \dots, I_N)$

Assuming speckle is zero-mean, independent, and stationary, the noise variance decreases by $\frac{1}{N}$. MLA produced stable backgrounds and preserved large echoes but blurred moving vessels due to temporal averaging. It outperformed RPCA and MTI, as shown in Figure 17, but underperformed relative to NLM.

NLM (Temporal Non-Local Means) identifies similar spatiotemporal patches $P(p)$ and computes the denoised value as:

$$\hat{I}(p) = \frac{1}{Z(p)} \sum_{q \in N(p)} w(p, q) \cdot I(q)$$

Where

$$w(p, q) = \exp\left(-\frac{|P(p) - P(q)|_2^2}{h^2}\right)$$

and $Z(p)$ is the normalization factor. NLM computes similarities in 3D (space-time) and averages based on those weights. It retains both moving and stationary objects by leveraging high-redundancy spatiotemporal structures. In Figure 17, NLM preserved low-RCS targets and minimized background clutter, outperforming all other multi-frame techniques.

Part 4c: HDR fusion

HDR fusion combines three radar sweeps—high-gain, medium-gain and low-gain—to capture both faint and strong echoes in a single image. The high-gain sweep brings out low-reflectivity targets (small boats, buoys), the low-gain sweep prevents saturation of bright features (ships, shorelines), and the medium-gain sweep bridges the two, reducing the radiometric gap. However, challenges arise due to motion misalignment and non-calibrated gain transitions.

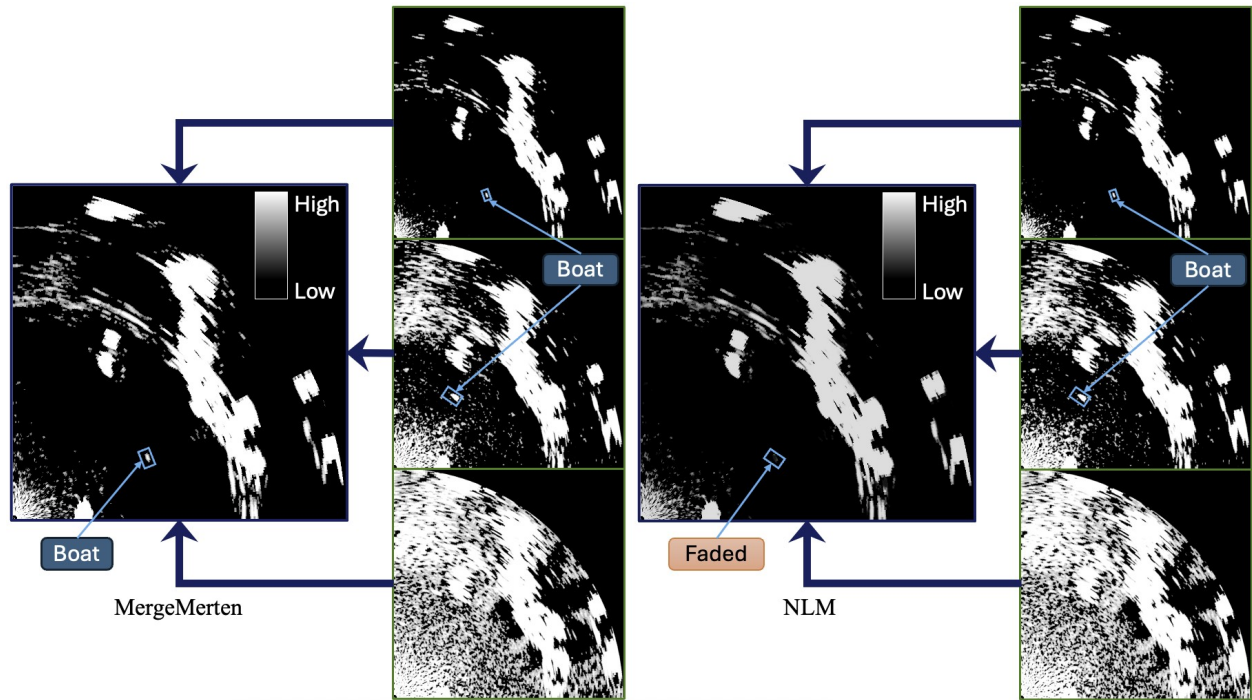


Figure 18: A comparison of Merge Merten and NLM(applied in a non-standard application) for HDR. NLM performs poorly for moving objects

The **Merge-Mertens exposure-fusion** algorithm constructs a weight map for each gain level based on local contrast, local variance (as a proxy for saturation in single-channel radar) and a “well-exposedness” term

$$E(p) = \exp \left[-\frac{(I(p) - 0.5)^2}{2\sigma^2} \right]$$

where $I(p)$ is the normalized pixel intensity and σ controls the emphasis on mid-range values. These three weight maps are normalized and blurred into Gaussian pyramids $G_i^{(l)}$; simultaneously, each gain image is decomposed into a Laplacian pyramid $L_i^{(l)}$. At each level l , the fused pyramid coefficient is

$$L_{\text{fused}}^{(l)}(p) = \sum_{i \in \{\text{hi, med, lo}\}} \frac{G_i^{(l)}(p) L_i^{(l)}(p)}{\sum_j G_j^{(l)}(p)}$$

and collapsing the fused Laplacian pyramid reconstructs the HDR image. By assigning near-zero weight to saturated or under-exposed pixels at every scale, Merge-Mertens achieves a smooth, ghost-tolerant blend without explicit sensor calibration.

In practice, as seen in Figure 18, Merge-Mertens successfully rendered both dim and bright echoes, but vessels that moved more than one beamwidth between gain captures left faint “ghost” outlines. These artifacts arise because even sub-second delays allow target displacement, which the multiscale blending cannot fully absorb.

An alternative attempt treated the three gain images as temporal frames and applied **temporal non-local means (NLM)** denoising. NLM seeks similar spatiotemporal patches $P(p)P(p)$ and computes

$$\hat{I}(p) = \frac{1}{Z(p)} \sum_q \exp\left(-\frac{|P(p) - P(q)|_2^2}{h^2}\right) I(q)$$

where h is the filtering bandwidth. Because the three gains differ by 20–40 dB and suffer non-linear log detection, patch distances become dominated by mean-level shifts rather than structural similarity. In addition, any motion between the captures causes patches of the same object to misalign. The result is indiscriminate averaging of unrelated content: coastlines smear, moving boats blur into gray halos, and weak targets disappear entirely.

Part 5: Data Storage and Formatting + Hardware Updates

Part 5a: Storage & Conversion

NetCDF is a self-describing, array-oriented standard with built-in named dimensions, rich metadata and chunked I/O. Any researcher can open a NetCDF file in Python, MATLAB, R or C and immediately understand its contents; units, long names, provenance and all; without separate schema files or custom parsers. In contrast, raw binaries (.bin), NumPy arrays (.npy) or MATLAB files (.mat) lack a universal metadata convention; flat text (.csv/.txt) requires sidecar descriptors; and folder-wide compressors (e.g. ZSTD) force full-file decompression even for small slices. The combination of portability, partial-read performance and community conventions (such as CF-metadata) establishes NetCDF as the de facto choice for long-term scientific data stewardship.

Raw radar sweeps arrive as CSV files; each row recording range, gain, angle and the time of the first sweep. Immediately, these CSVs are ingested into a two-level NetCDF structure via a three-stage Rust pipeline that performs deduplication, sanitization and conversion into a compressed Echo variable. Front-loading NetCDF ensures that every downstream tool (MATLAB’s ncread, Python’s netCDF4/xarray, R’s ncdf4) reads the same self-describing, chunk-friendly archive, eliminating bespoke CSV parsers and accelerating both interactive analysis and automated workflows.

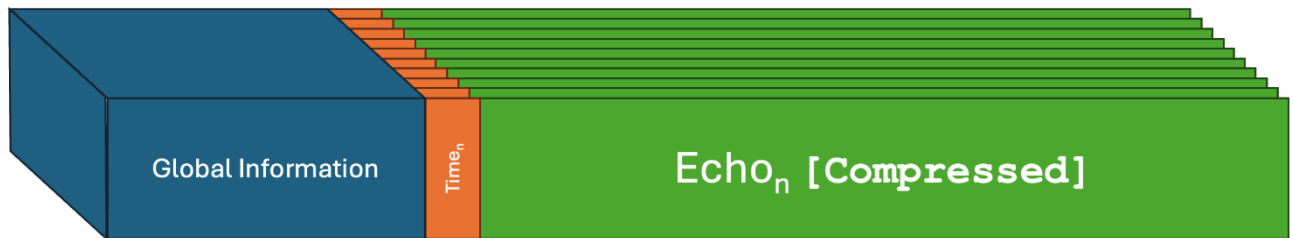


Figure 19: High-Level overview of NetCDF data storage. Global data contains location and radar settings.

At the top level(Figure 19), each NetCDF file opens with a set of global attributes, radar latitude and longitude, transmitter gain, start-of-sweep timestamp, and so on, followed by an unlimited “time” dimension. Defining time as an unlimited axis makes it possible to append future sweeps without rewriting the file. For every index along that axis, the two-dimensional Echo(gates, angle) variable is stored and compressed in place using DEFLATE-9. Because all metadata lives in the header, (i.e. units, long names, provenance) tools like MATLAB’s ncread, Python’s netCDF4/xarray or R’s ncdf4 can open the file and immediately interpret every field without sidcar schemas or manual parsing.

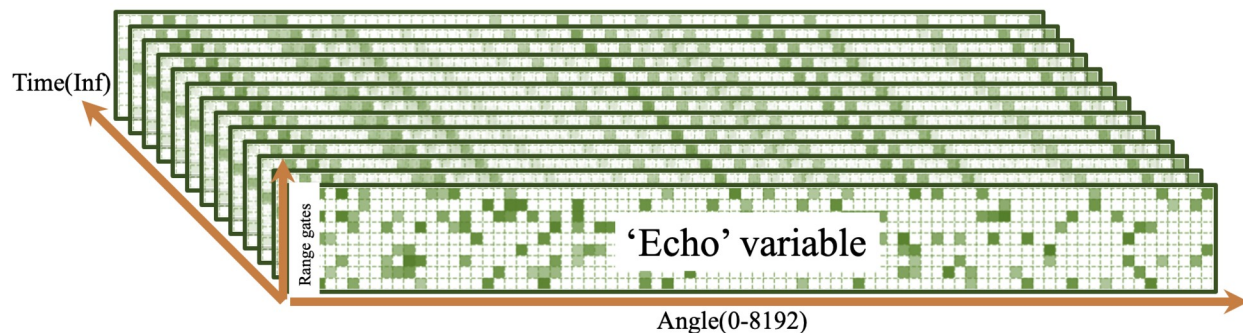


Figure 20: Lower-level look at NetCDF structure. Scaled at 1:100

Drilling down (Figure 20), the Echo array sits on two dimensions; time (unlimited length) and angle (0–8192 steps). The NetCDF4’s DEFLATE 9 filter (with byte-shuffle) was used to greatly reduce storage size. Crucially, NetCDF’s chunked I/O lets readers fetch a single time-slice or angular subset without decompressing the entire archive.

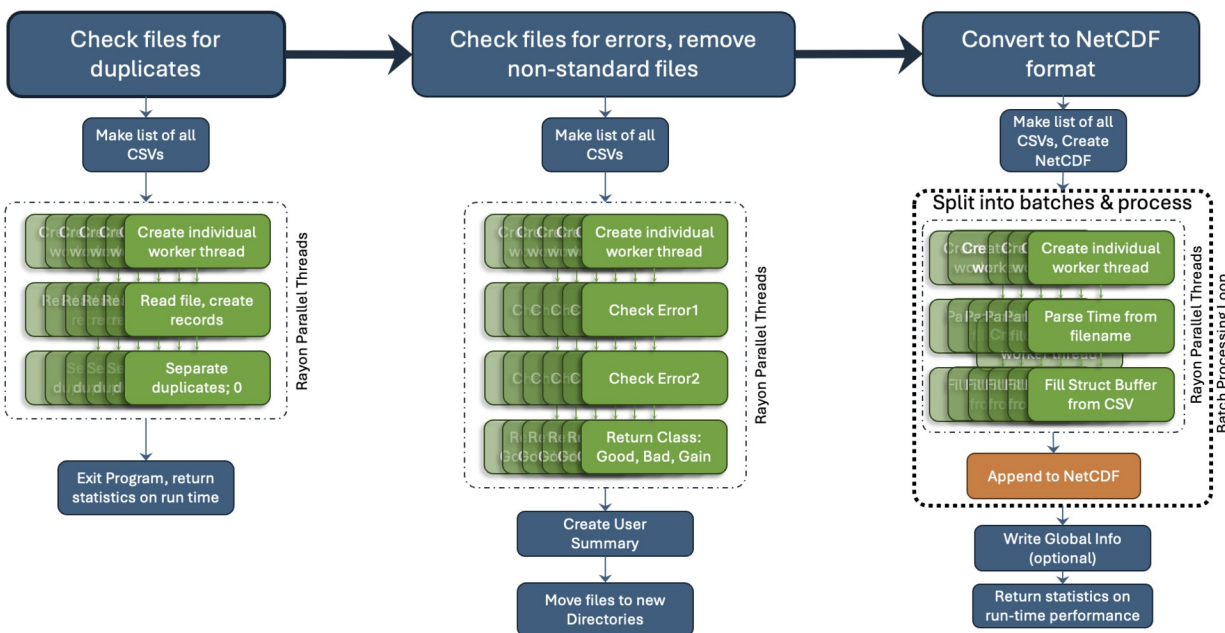


Figure 21: RUST processing pipeline for radar data. Converts CSV to NetCDF

To create the NetCDF4 files from massive datasets (>0.2 TB), a highly-parallelized Rust program (Figure 21) spins up a work queue and three stages of workers:

1. **Deduplication**, where a coordinator thread lists new CSVs and dispatches them to cleaners that drop exact-duplicate records.

2. **Sanitization**, in which validators normalize data types, enforce $0 \leq \text{angle} < 360^\circ$ (8192), and quarantine any corrupt files.
3. **Conversion**, where threads batch-read cleaned CSV rows into memory structs and append them directly into the NetCDF file, then write out the global attributes all at once.

Running all stages in parallel across multiple cores real run-time low; even though compression is CPU-heavy, the work is amortized over each batch and scales nearly linearly with core count.

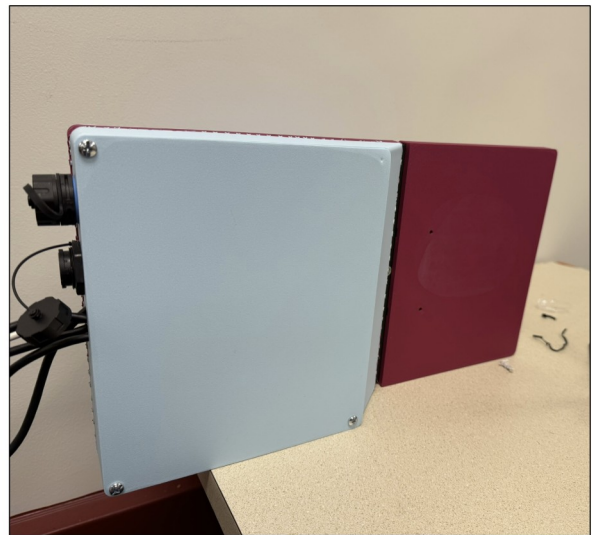
Once the canonical NetCDF is complete, simple Python or MATLAB scripts slice each time-step into 16-bit PNG frames and assemble an MP4 via FFmpeg. By standardizing on NetCDF, we eliminated dozens of bespoke CSV parsers, gained self-describing metadata, random-access subsetting, and cross-language interoperability, all outweighing the raw speed/size gains of a one-off ZSTD archive.

Part 5b: Capture-Rig Refinements

Field deployment was dramatically simplified by replacing the ad-hoc open-frame prototype (Figure 22, left) with a purpose-built sealed enclosure and elevation mount (Figure 22, right). The new housing protects sensitive electronics from moisture and debris, streamlines cable management, and locks down the antenna at a fixed height, reducing rig-up time from several minutes of tinkering with software to under one minute of plug-and-play.



OLD



NEW

Figure 22: Original open-frame prototype (left) versus the new sealed enclosure with finalized elevation mount (right), offering faster setup and better environmental protection.

Beyond the enclosure itself, the entire lakeside transport and mounting system was reimagined. As shown in Figure 23, the original rig relied on hand-carried tripods and loose power packs (left), whereas the proposed design integrates all components onto a compact wheeled platform (right). The sealed dome sits atop a sturdy three-leg elevation apparatus, with battery, computer, and human-interface unit fastened inside. A dedicated transportation device allows a single operator to roll the rig into position and begin data capture immediately, eliminating the need for repeated assembly and disassembly on uneven terrain.

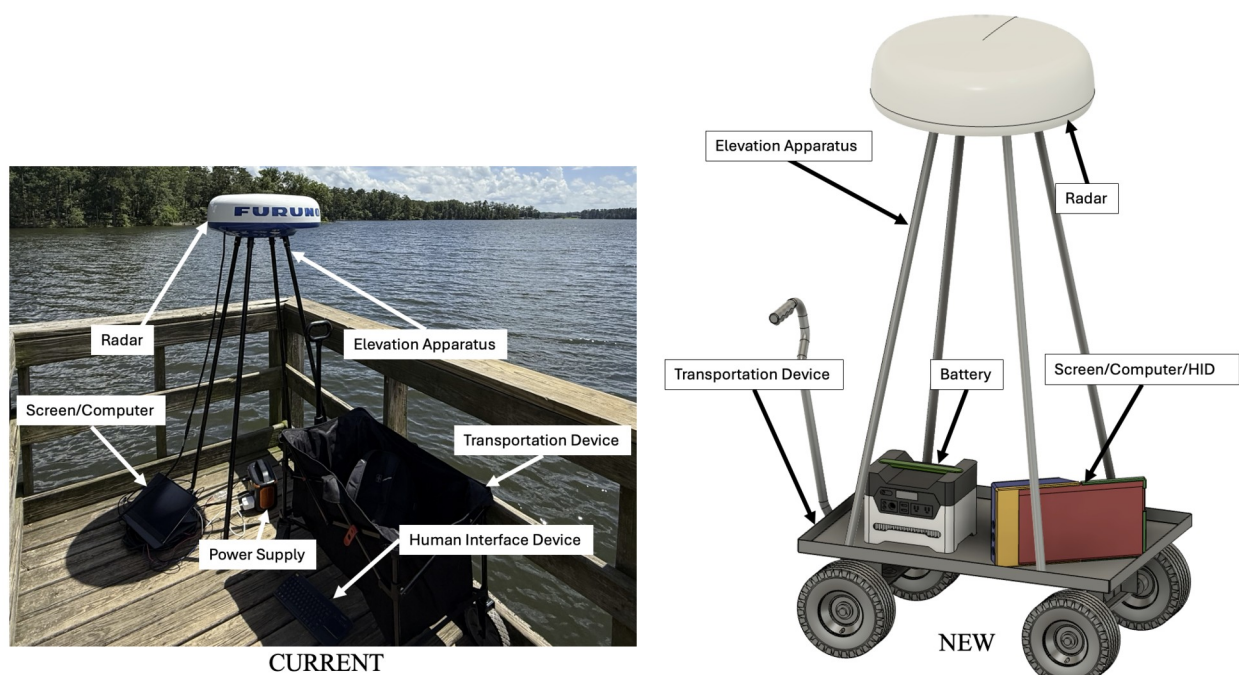


Figure 23: Existing open-frame lakeside rig (left) versus the new wheeled platform (right) with sealed dome, integrated elevation apparatus, power supply, transportation device, and human-interface unit

Part 5c: Collection Software Improvements

Bringing the boot time down from several minutes to under twenty seconds began with swapping out the bloated Raspberry Pi OS image for a leaner build (while still retaining GUI support). By stripping unnecessary services and drivers, the Pi now reaches a useable prompt in under 20 s, compared with four to six minutes on the previous install.

Simultaneously, the data capture code was refactored and ported from C into modern C++. The main program shrank from roughly 2,000 lines to just over 500, achieved by removing redundant function calls, consolidating I/O routines, and leveraging STL containers and RAII for resource management. The result is a far more modular codebase, where the boost web server, radar SDK worker, calibration handler, sweep sequencer, and file writer threads each live in their own component.

Figure 24 illustrates this multithreaded workflow: after initialization, three core modules spin up in parallel, while dedicated threads manage range/gain sequencing and raw data/PNG writes. Pushing so many concurrent threads, five in the main pipeline plus any callback threads from the radar SDK, has begun to bump against the Pi 5's four-thread architectural limit, leading to scheduler contention under peak load, resulting in severe data loss when adding additional processing threads (i.e. edge processing). Future work will explore rebalancing responsibilities to avoid thread starvation or implementing more 'offline' processing.

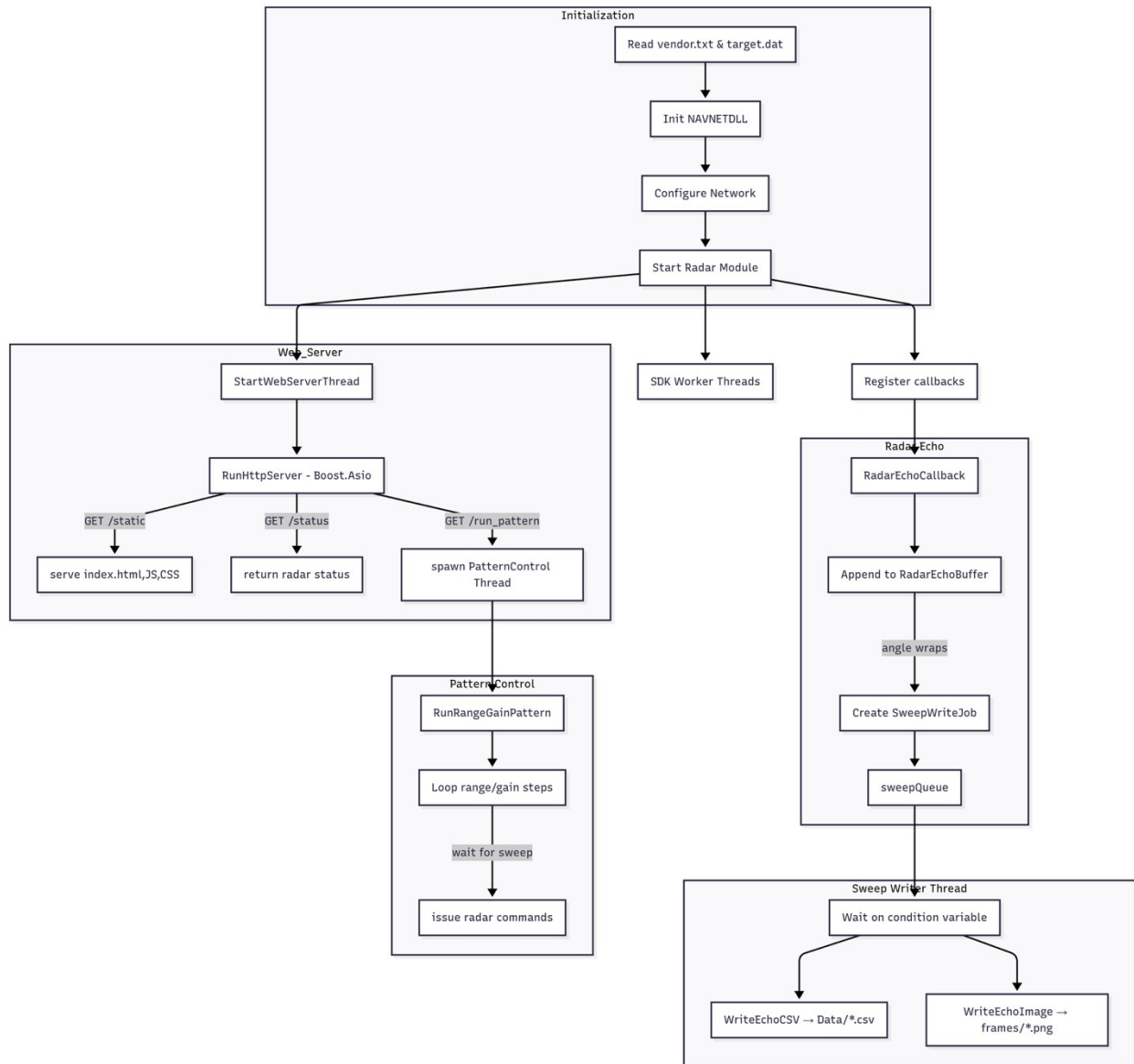


Figure 24: Block diagram of the capture software's multithreaded workflow: after initialization, three core modules run in parallel while dedicated threads sequence range/gain sweeps and write raw data plus PNG frames

Part 5d: Calibration Sequencing

Calibration proceeds in two tiers. As seen in Figure 25, the mini calibration varies five discrete ranges (0.0625, 0.125, 0.25, 0.50 and 0.75 NM) and 100 gain settings (0–99), capturing five back-to-back sweeps at each gain (discarding the first as a warm-up). With sea- and rain-clutter disabled, this produces 2,500 valid sweeps in one run. In high-RPM mode, a mini calibration completes in about 40 minutes; in low-RPM it takes roughly 80 minutes.

For thorough system characterization, the full-calibration adds two interpolation modes and four RezBoost settings (2×4 additional configurations), multiplying the sweep count to 20,000. At high RPM

this requires ~ 320 minutes (≈ 5 h 20 m), and at low RPM ~ 640 minutes (≈ 10 h 40 m). The resulting data feed into automated routines that compute gain-vs-range response curves, quantify ADC linearity, and generate lookup tables for real-time gain scheduling on the radar web server.

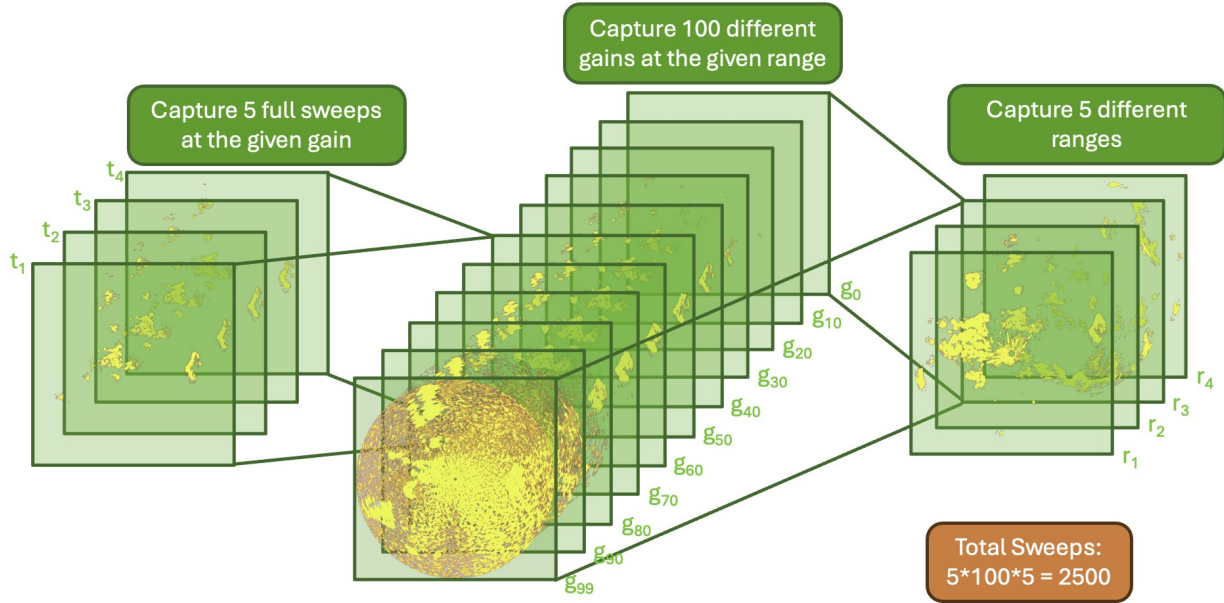


Figure 25: Nested calibration workflow showing five range settings (r_1 – r_5), 100 gain steps (0–99), and five sweeps per gain—yielding $5 \times 100 \times 5 = 2,500$ sweeps per mini-calibration pass

Part 6: Self-Supervised Temporal Exposure MAE Prototype

Part 6a: The Idea

Raw dual-pulse radar sweeps are reshaped into a four-dimensional tensor, range $\times$ azimuth $\times$ time/gain $\times$ pulse, and fed into a Masked Autoencoder that learns both motion and exposure priors without any manual labels. In the first stage, 75 % of the voxels in each current frame are randomly masked and reconstructed conditioned on the previous time slice; this forces the encoder to internalize object trajectories and temporal consistency (see Figure 26 and Figure 27).

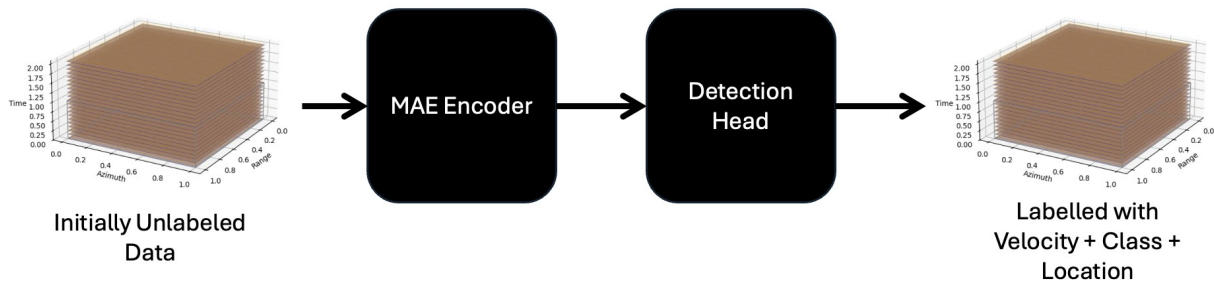


Figure 26: Complete proposed MAE detection pipeline.

The second stage applies an exposure-aware mask, borrowing from SelfHDR, to treat clipped or saturated low-gain voxels as missing data, compelling the network to fuse in high-gain returns where needed. Combined, these self-supervised tasks teach the model to handle both dynamic-range extremes and moving targets on low-bit, non-Doppler radar hardware.

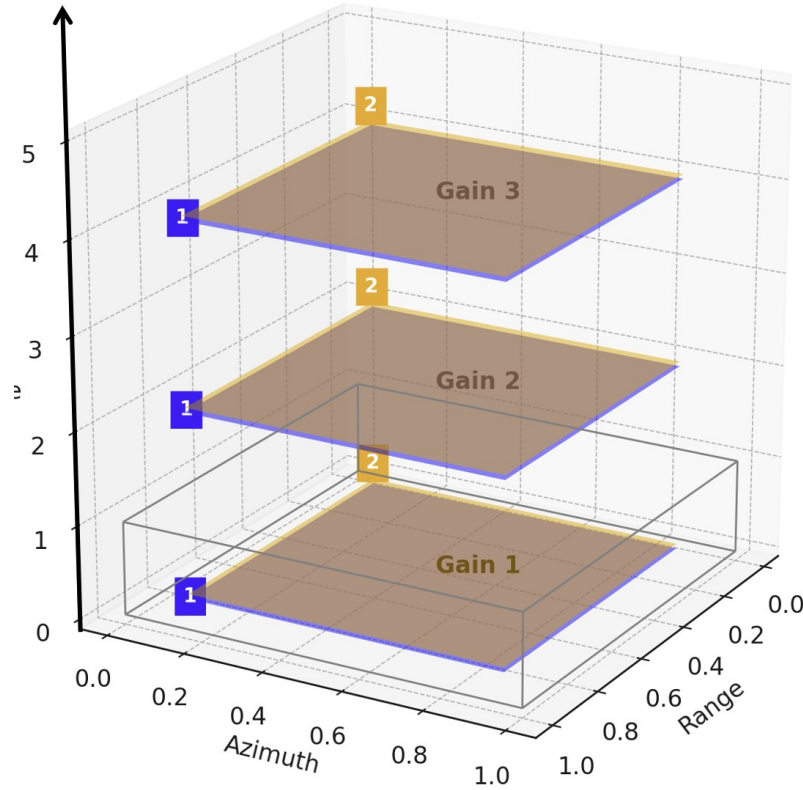


Figure 27: Visual representation of the 4D Radar Data Tensor the MAE is trained on

Part 6b: Results

With just 132 frames for training, 40 epochs of training at batch size 1, and an L_2 +SSIM reconstruction loss, the Temporal MAE captured some broad background structure and slow object motion but struggles to resolve most targets. Reconstruction MSE remained high and SSIM on masked regions produced blurred or incomplete restorations. The exposure-aware pass improved contrast recovery in saturated areas but introduced edge artifacts, indicating that a learnable or adaptive masking threshold may be required.

Part 7 Future

Part 7a: Algorithms and Model Development

The first priority will be to finalize the optimal S1/S2 gain values so that the radar can operate in full HDR mode, ensuring maximal dynamic-range capture across all scenarios. Once the gain window is locked in, work will shift to completing the preliminary HDR-to-object-detection pipeline, whereby high-, medium- and low-gain frames are fused and fed directly into a lightweight detection network. In parallel, the Masked Autoencoder prototype will undergo further testing in both its temporal and exposure-aware configurations, leveraging a larger and

more varied training corpus plus adaptive masking schedules to improve reconstruction fidelity and increase detection accuracy.

Part 7b: Experiments and Testing

The calibration web-server needs to be enhanced to support a truly end-to-end mini- and full-calibration run; in addition to range and gain selectors, users will need controls for STC, pulse length, RezBoost, rotation speed and interpolation, and the underlying threading model must be refactored to eliminate dropped sweeps. Once the server is robust, system-level field trials will validate both one-click calibration and HDR-detection performance under varied weather conditions. If capacity allows, undergraduate-led efforts will integrate IMU/GPS data into radar headers for precise georeferencing, add a synchronized RGB camera for multimodal object-labelled datasets, and build a simple touch-accessible frontend so operators can deploy the entire capture, calibration and HDR pipeline from a single panel.